

TECHNICAL BULLETIN**DIXIE CHEMICAL**

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ECA 300D
CAS Reg. No 25550-51-0, 34090-76-1

Typical Properties

Properties	Value	Test Method
Molecular Weight (average)	168	
MHHPA, %	87.0-93.0 Min	D-160
Maleic Anhydride, %	0.1 Max	D-160
Acid, %	2.0 Max	D-160B3
Viscosity, cPs	45-65	D-150F
Specific Gravity @ 25°C, g/cc	1.100-1.200	D-102 (ASTM D4052)
Appearance	Clear Liquid	D-104 (Visual)
Volatiles, ppm	100 Max	D-160A
Color, Pt-co	1 Max	D-150G
Freezing point, °C	-29 to -31	ASTM D1015
Pour point, °F	-38 to -40	ASTM D97

Applications

ECA 300D is a clear colorless, high purity liquid which can be used alone or in combination with hexahydrophthalic anhydride to produce a lower melting mixture or with other anhydrides to modify physical properties. Its low residual unsaturation level gives maximum resistance to yellowing in coatings and castings, making it desirable where premium optical and

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electrical performance are needed in epoxy casting and encapsulation applications. Also, gives excellent cure response at cure temperatures, resulting in high crosslinking and relatively high Tg.

For more information on the use of anhydrides like ECA 300D as epoxy curing agents, please consult the Technical Bulletin, "Formulating Anhydride-Cured Epoxy Systems", available from Dixie Chemical.

ECA 300D was formulated with a Standard BPA Liquid Epoxy Resin. The sample was formulated with 100 parts of epoxy resin, 83 parts ECA 300D, and 1 part 1-Methylimidazole. Viscosity was measured at 25°C, and gel time was measured at 100°C. Samples were cured for 1 hour at 120°C, followed by a post-cure of 1 hour at 220°C, and glass transition temperatures were measured using a differential scanning calorimeter. The following table summarizes these results:

Epoxy type	ECA 300D, phr	Viscosity @ 25 °C, cP	Gel Time ¹ , min	Tg ² , °C
Standard BPA Liquid Epoxy	83	1200	35	156

1. Shyodu Hot Pot (100g @ 100°C)

2. Cured 1 hr @ 120°C and 1 hr @ 220°C. Analyzed in DSC: 40°C/min to 250°C

Handling & Storage

ECA 300D will react with water to form methylhexahydrophthalic acid. This is normally undesirable, so ECA 300D should be stored in such a way that it is carefully protected from moisture contamination.

For more details on the design of bulk storage for ECA 300D, consult the Dixie Chemical Company brochure "Epoxy Curing Agent Storage Requirements."

In some cases ECA 300D may crystallize. If crystallization is observed, the material can be heated to return the product to a liquid state.

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Loosen bung caps on each drum and place the drums in a hotbox or oven at 80°C for 24 hours. Upon removal from the hotbox, the product should be free of crystals. The product should be used within 48 hours of melting to ensure no reoccurrence of crystal formation.

Heating ECA 300D following this procedure does not alter the chemical make-up or degrade any properties of the product.

Health Hazards

Read and understand the relevant Safety Data Sheets (SDS) for all products before use. These anhydrides are primary skin and eye irritants. Avoid contact with skin, eyes, and clothing. Use only with adequate ventilation. In case of contact, follow the procedures outlined in the SDS. Generally, these procedures include immediately flushing the affected skin or eyes with copious amounts of water for at least 15 minutes. In the case of eye contact, get medical attention. Wash contaminated clothing before reuse.

Follow the recommendations in the SDS for personal protective equipment when handling these materials. At a minimum, these procedures typically include protective chemical goggles, impenetrable gloves, and measures to avoid breathing chemical vapors.

Availability

ECA 300D is available from Dixie Chemical in Pasadena, Texas. Contact your Dixie representative for details.